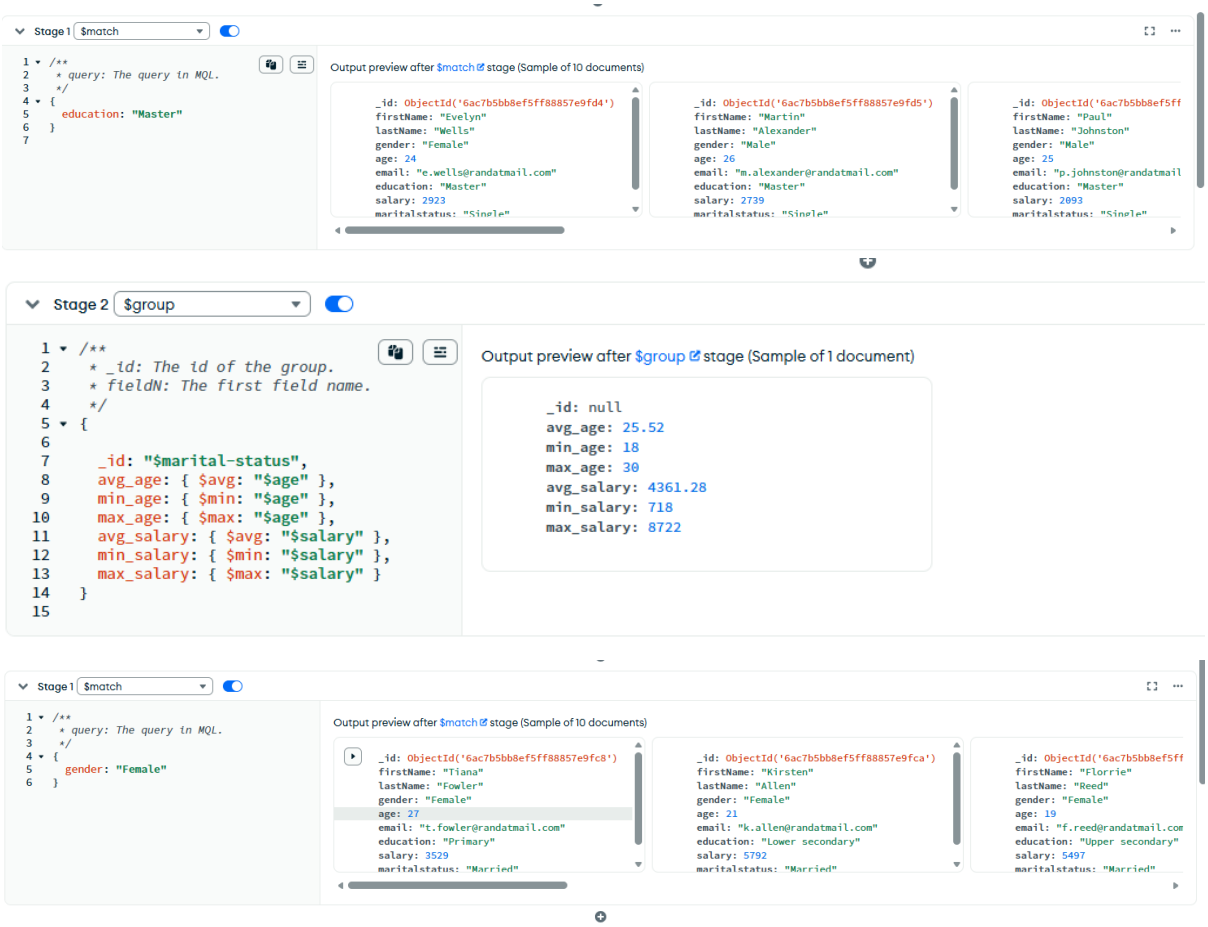
WEEK 2

Reflection:

In this week lab I practiced with MongoDB Atlas aggregation queries. Firstly, I wrote an aggregation to compute the average, minimum, and maximum age and salary for people with a Master's education, grouped by marital status. Secondly, I wrote an aggregation to compute the minimum, maximum, and average salaries of females grouped by age. Thirdly I wrote another aggregation to compute these same values for males. Finally, I grouped the data by gender and marital status, to compute the total number of married and unmarried males and females, using the $group and $sum operators. This proved to be a good way of developing my skills with $match, $group, $avg, $min and $max. I am now more confident in writing aggregation queries and understanding their results. Attached is a screenshot of each of the four queries and their results as evidence.



Stage 2 $group

```
/**
 * _id: The id of the group.
 * fieldN: The first field name.
 */
{
  _id: "$age",
  min_salary: { $min: "$salary" },
  max_salary: { $max: "$salary" },
  avg_salary: { $avg: "$salary" }
}
```

Output preview after $group stage (Sample of 10 documents)

```
_id: 26
min_salary: 509
max_salary: 9611
avg_salary: 5319.5
```

```
_id: 22
min_salary: 3565
max_salary: 9925
avg_salary: 5782
```

Stage 1 $match

```
/**
 * query: The query in MQL.
 */
{
  gender: "Male"
}
```

Output preview after $match stage (Sample of 10 documents)

```
_id: ObjectId('6ac7b5bb8ef5ff88857e9fc6')
firstName: "Justin"
lastName: "West"
gender: "Male"
age: 27
email: "j.west@randatmail.com"
education: "Doctoral"
salary: 5783
maritalstatus: "Married"
```

```
_id: ObjectId('6ac7b5bb8ef5ff88857e9fc7')
firstName: "Daryl"
lastName: "Johnson"
gender: "Male"
age: 20
email: "d.johnson@randatmail.com"
education: "Upper secondary"
salary: 4450
maritalstatus: "Married"
```

```
_id: ObjectId('6ac7b5bb8ef5ff
firstName: "Alen"
lastName: "Barnes"
gender: "Male"
age: 26
email: "a.barnes@randatmail.c
education: "Upper secondary"
salary: 6332
maritalstatus: "Married"
```

Stage 2 $group

```
/**
 * _id: The id of the group.
 * fieldN: The first field name.
 */
{
  _id: "$age",
  min_salary: { $min: "$salary" },
  max_salary: { $max: "$salary" },
  avg_salary: { $avg: "$salary" }
}
```

Output preview after $group stage (Sample of 10 documents)

```
_id: 30
min_salary: 1260
max_salary: 9989
avg_salary: 5363.090909090909
```

```
_id: 19
min_salary: 1221
max_salary: 9543
avg_salary: 5469.75
```

Stage 1 $group

```
/**
 * _id: The id of the group.
 * fieldN: The first field name.
 */
{
  _id: {
    gender: "$gender",
    marital_status: "$marital-status"
  },
  count: { $sum: 1 }
}
```

Output preview after $group stage (Sample of 2 documents)

```
▸ _id: Object (1)
  count: 93
```

```
▸ _id: Object (1)
  count: 108
```